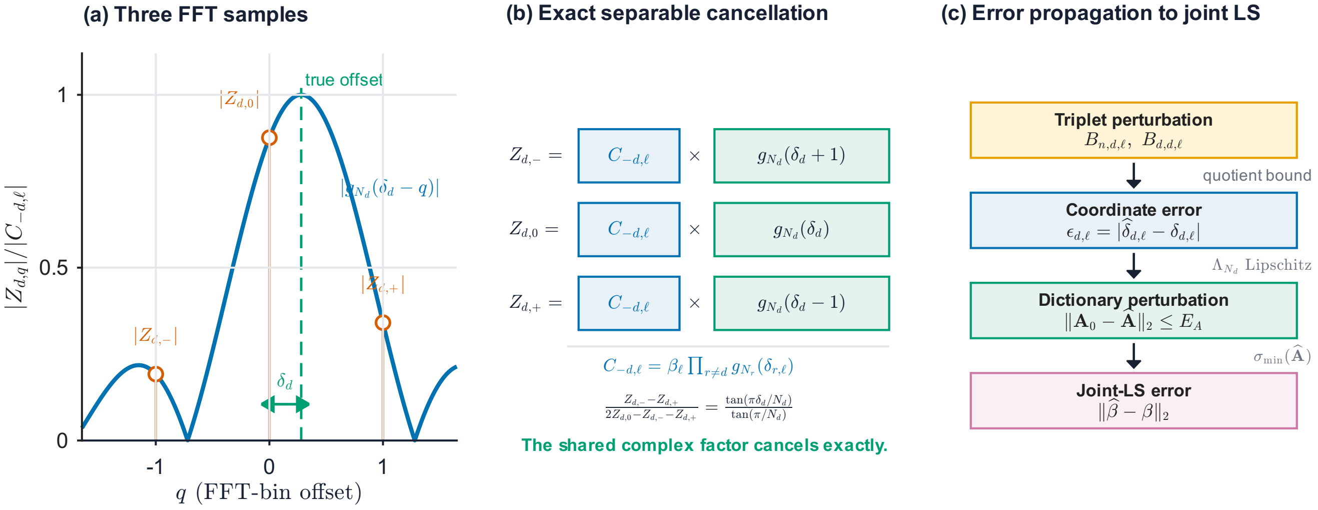
(a) Three FFT samples
true offset
$|Z_{d,0}|$
$|g_{N_d}(\delta_d - q)|$
$|Z_{d,+}|$
$|Z_{d,-}|$
δ_d
$|Z_{d,q}|/|C_{-d,\ell}|$
q (FFT-bin offset)
(b) Exact separable cancellation
$Z_{d,-} = C_{-d,\ell} \times g_{N_d}(\delta_d + 1)$
$Z_{d,0} = C_{-d,\ell} \times g_{N_d}(\delta_d)$
$Z_{d,+} = C_{-d,\ell} \times g_{N_d}(\delta_d - 1)$
$C_{-d,\ell} = \beta_\ell \prod_{r \neq d} g_{N_r}(\delta_{r,\ell})$
$\frac{Z_{d,-} - Z_{d,+}}{2Z_{d,0} - Z_{d,-} - Z_{d,+}} = \frac{\tan(\pi\delta_d/N_d)}{\tan(\pi/N_d)}$
The shared complex factor cancels exactly.
(c) Error propagation to joint LS
Triplet perturbation
$B_{n,d,\ell}, \; B_{d,d,\ell}$
quotient bound
Coordinate error
$\epsilon_{d,\ell} = |\widehat{\delta}_{d,\ell} - \delta_{d,\ell}|$
Λ_{N_d} Lipschitz
Dictionary perturbation
$\|\mathbf{A}_0 - \widehat{\mathbf{A}}\|_2 \leq E_A$
$\sigma_{\min}(\widehat{\mathbf{A}})$
Joint-LS error
$\|\widehat{\beta} - \beta\|_2$